# Toward an Efficient General-Purpose Time-Series Model

**Progress report: TimeMixer, TimeMixer++, SparseTSF and SparseTSFPlus on the TimeMixer++ task suite**

23 October 2026 · prepared for the biweekly supervision meeting

> Every number in the tables of this report is generated from the per-run logs in the public repository (`BenguBarisBalkan/efficient-general-purpose-timeseries-models`) by `TimeMixer/benchmarks/tools/build_progress_document.py`; none is typed by hand. **Local** numbers were measured on the machine described in Section 1.3 under one shared protocol. **Published** numbers are copied from the TimeMixer and TimeMixer++ papers, were produced on other hardware with the authors' own tuning, and are shown in *italics* in separate columns. The two are never mixed in one cell.


## Abstract

This thesis asks how efficient general-purpose time-series models are, and whether a model can be built that is both efficient and capable across the eight task families of the TimeMixer++ benchmark: long-term and short-term forecasting, imputation, few-shot and zero-shot forecasting, anomaly detection and classification. We compare three published models: TimeMixer (ICLR 2024), TimeMixer++ (ICLR 2025; reimplemented from the paper, since no official code exists) and SparseTSF (ICML 2024, about 1,000 parameters). We then build **SparseTSFPlus**, which starts from SparseTSF and adds only components that measurably earn their parameters.

Over the last two weeks the comparison went from SparseTSF-only on most tasks to all tasks run locally for TimeMixer and SparseTSF. Sixteen ideas and checks for the SparseTSF backbone were tested, each with a prediction recorded before its runs. The result is one fixed SparseTSFPlus configuration, with 37 to 8,661 parameters outside classification, that beats SparseTSF beyond run-to-run noise on five tasks over three seeds:

| Task | Gain over SparseTSF (3 seeds) |
|---|---|
| PEMS | −24% |
| Imputation | −16% |
| M4 | −6% |
| Few-shot | −1.4% |
| Long-term | −1.0% |

On every forecasting task it is within 0.7% of choosing components per task. It uses 1.35× the energy of SparseTSF and about one fourteenth of TimeMixer's. Against TimeMixer it is ahead on few-shot and zero-shot transfer, and behind on long-term, M4, PEMS, imputation and classification. Repeating runs over seeds also showed that several single-seed conclusions in imputation and anomaly detection were noise, and those are corrected here.


## 1. Introduction

### 1.1 Research question

General-purpose time-series models, sometimes called time-series pattern machines, use one architecture for many task families. They are trained per task and dataset; they are not pretrained foundation models such as TimesFM or Chronos, although the zero-shot task probes a similar property. The state of the art in this family, TimeMixer++, reaches strong accuracy with a multi-scale, attention-based backbone; in this study TimeMixer and TimeMixer++ use $10^4$ to $10^7$ parameters. The question of this thesis is how much of that capacity is needed: **can a**

**model of a few hundred to a few thousand parameters be made capable across all eight tasks, with one configuration?**

### 1.2 Models

- **TimeMixer** (Wang et al., ICLR 2024). Decomposes each series into seasonal and trend parts at several temporal scales and mixes them across scales with MLPs. Upstream code; run locally with its published scripts where they exist (long-term, M4, PEMS), otherwise with its own long-term architecture.
- **TimeMixer++** (Wang et al., ICLR 2025). Adds multi-resolution "time imaging" by FFT-selected periods, dual-axis attention, and multi-scale mixing. No official code is public, so this study reimplemented it from the paper. The reimplementation is numerically unstable at the paper's learning rate and was only stable at `lr = 5e-5`, which undertrains it.
- **SparseTSF** (Lin et al., ICML 2024). Folds each channel's input into a grid of *phase × period count* and applies one small linear map across periods, so a whole forecaster costs tens to thousands of parameters. Published for long-term forecasting only; extended here to all task families (Section 3.1).
- **SparseTSFPlus** (this work). SparseTSF plus individually switchable components ("units"), each defaulting to off. With every unit off it is bitwise identical to SparseTSF, which an automated check verifies after every change.

### 1.3 Protocol and setup

- **Tasks, datasets and metrics** follow the TimeMixer++ paper. Deviations are listed in Section 2.3.
- **Hardware:** one Windows laptop with a 4 GB NVIDIA GPU (RTX 3050 Ti) and 16 GB of RAM. Batch sizes are reduced from the papers' values where memory requires.
- **Hyperparameters:** each local number uses the model's own published settings where they exist. Every SparseTSFPlus variant copies the settings of the run that produced the SparseTSF number it is compared with, so the two differ only in the units.
- **Selection** of units uses validation loss, never test loss. Imputation, anomaly detection and classification early-stop on the test split in the upstream code shared by all models; their results are therefore a sensitivity analysis.
- **Seeds:** every run uses seed 2021 unless stated otherwise. The headline comparisons were repeated with seeds 2022 and 2023 (Section 4.2). A difference counts as a *finding* only if it has the same sign on all three seeds and its mean exceeds twice the standard deviation of the per-seed differences.

## 2. Three-model comparison

Each table reports one value per dataset. The columns named after a model are **local**; the columns marked " (paper)" are **published** values in *italics*. The best local value per row is in **bold**; published values are never bold. "partial (k/4)" marks a dataset where only k of its four cells finished, so no mean is reported. "—" means not run (local) or not reported (published).

Metrics, by task group:

- **Long-term:** MSE, mean over horizons 96, 192, 336 and 720 with a 96-step input.
- **M4:** OWA over the six subsets.
- **PEMS:** MAE, input 96 → output 12.
- **Imputation:** masked MSE, mean over mask rates 12.5–50%, input 96.
- **Few-shot:** MSE with 10% of the training data, mean over the four horizons.
- **Zero-shot:** MSE when training on dataset A and testing on B, mean over the four horizons.
- **Anomaly detection:** point-adjusted F1 (%).

- **Classification:** accuracy (%).

Lower is better (↓) except for F1 and accuracy (↑).

### 2.1 State at the start of the reporting period (11 October 2026)

Table 1 shows every result that existed on 11 October 2026. Each log is read from git at commit `72040e6`, so a cell counts only if its run had finished by then. At that point only SparseTSF had been run on the per-task benchmarks. TimeMixer had been run locally only on long-term forecasting and on five imputation datasets, and TimeMixer++ only on long-term forecasting. Everything else in the TimeMixer and TimeMixer++ columns is published.

**Table 1.** Three-model comparison on 11 October 2026 (commit `72040e6`).

| Dataset | TimeMixer | TimeMixer++ | SparseTSF | TimeMixer (paper) | TimeMixer++ (paper) |
|---|---|---|---|---|---|
| **Long-term (MSE ↓)** | | | | | |
| ETTh1 | 0.4578 | 0.7467 | **0.4347** | *0.4465* | *0.4185* |
| ETTh2 | 0.3730 | 0.4507 | **0.3566** | *0.3648* | *0.3390* |
| ETTm1 | **0.3841** | 0.4579 | 0.4172 | *0.3812* | *0.3685* |
| ETTm2 | **0.2799** | 0.3125 | 0.2884 | *0.2752* | *0.2687* |
| Weather | **0.2454** | 0.2756 | 0.2741 | *0.2403* | *0.2263* |
| Solar-Energy | 0.2319 | partial (1/4) | — | *0.2162* | *0.2032* |
| Electricity | 0.1830 | — | — | *0.1822* | *0.1645* |
| Traffic | partial (2/4) | — | — | *0.4848* | *0.4158* |
| **M4 (OWA ↓)** | | | | | |
| M4, 6 subsets (OWA) | — | — | 0.972 | *0.840* | *0.821* |
| **PEMS (MAE ↓)** | | | | | |
| PEMS03 | — | — | 28.28 | — | — |
| PEMS04 | — | — | 36.12 | — | — |
| PEMS07 | — | — | 42.89 | — | — |
| PEMS08 | — | — | 29.25 | — | — |
| Average | — | — | 34.13 | *17.41* | *15.91* |
| **Imputation (MSE ↓)** | | | | | |
| ETTh1 | **0.1201** | — | 0.1633 | — | — |
| ETTh2 | **0.0664** | — | 0.0703 | — | — |
| ETTm1 | **0.0470** | — | 0.0715 | — | — |
| ETTm2 | **0.0278** | — | 0.0327 | — | — |
| Weather | **0.0315** | — | 0.0344 | — | — |
| Electricity | — | — | 0.1304 | — | — |
| **Few-shot (MSE ↓)** | | | | | |
| ETTh1 | — | — | 0.5517 | — | — |
| ETTh2 | — | — | 0.4297 | — | — |
| ETTm1 | — | — | 0.4935 | — | — |
| ETTm2 | — | — | 0.2885 | — | — |
| ETT average | — | — | 0.4408 | *0.4530* | *0.3960* |
| Weather | — | — | 0.2708 | *0.2420* | *0.2410* |
| Electricity | — | — | partial (3/4) | *0.1870* | *0.1680* |
| **Zero-shot (MSE ↓)** | | | | | |

| Dataset | TimeMixer | TimeMixer++ | SparseTSF | TimeMixer (paper) | TimeMixer++ (paper) |
|---|---|---|---|---|---|
| ETTh1 → ETTh2 | — | — | 0.3596 | *0.4270* | *0.3670* |
| ETTh1 → ETTm2 | — | — | 0.3207 | *0.3610* | *0.3010* |
| ETTh2 → ETTh1 | — | — | 0.4527 | *0.6790* | *0.5110* |
| ETTm1 → ETTh2 | — | — | 0.3958 | *0.4520* | *0.4170* |
| ETTm1 → ETTm2 | — | — | 0.2955 | *0.3290* | *0.2910* |
| ETTm2 → ETTm1 | — | — | 0.5329 | *0.5540* | *0.4270* |
| **Anomaly (F1 % ↑)** | | | | | |
| PSM | — | — | 95.26 | — | *97.60* |
| SMD | — | — | 80.02 | — | *86.50* |
| MSL | — | — | — | — | *85.82* |
| SMAP | — | — | — | — | *73.10* |
| SWaT (not obtainable) | — | — | — | — | *94.64* |
| **Classification (accuracy % ↑)** | | | | | |
| EthanolConcentration | — | — | 29.28 | — | — |
| FaceDetection | — | — | 68.59 | — | — |
| Handwriting | — | — | 8.59 | — | — |
| Heartbeat | — | — | 54.63 | — | — |
| JapaneseVowels | — | — | 61.62 | — | — |
| PEMS-SF | — | — | 84.39 | — | — |
| SelfRegulationSCP1 | — | — | 82.25 | — | — |
| SelfRegulationSCP2 | — | — | 52.78 | — | — |
| SpokenArabicDigits | — | — | 94.04 | — | — |
| UWaveGestureLibrary | — | — | 72.81 | — | — |
| Average | — | — | 60.90 | — | *75.90* |

## 2.2 Current state (23 October 2026)

Over the reporting period TimeMixer was run locally on every task, SparseTSF on the three remaining long-term datasets and on MSL and SMAP, and the two remaining imputation and few-shot cells were completed. Table 2 is the comparison as it stands; Table 3 counts the finished runs then and now.

**Table 2.** Three-model comparison on 23 October 2026.

| Dataset | TimeMixer | TimeMixer++ | SparseTSF | TimeMixer (paper) | TimeMixer++ (paper) |
|---|---|---|---|---|---|
| **Long-term (MSE ↓)** | | | | | |
| ETTh1 | 0.4578 | 0.7467 | **0.4347** | *0.4465* | *0.4185* |
| ETTh2 | 0.3730 | 0.4507 | **0.3566** | *0.3648* | *0.3390* |
| ETTm1 | **0.3841** | 0.4579 | 0.4172 | *0.3812* | *0.3685* |
| ETTm2 | **0.2799** | 0.3125 | 0.2884 | *0.2752* | *0.2687* |
| Weather | **0.2454** | 0.2756 | 0.2741 | *0.2403* | *0.2263* |
| Solar-Energy | **0.2319** | partial (1/4) | 0.3858 | *0.2162* | *0.2032* |
| Electricity | **0.1830** | — | 0.2218 | *0.1822* | *0.1645* |
| Traffic | partial (2/4) | — | 0.6346 | *0.4848* | *0.4158* |
| **M4 (OWA ↓)** | | | | | |
| M4, 6 subsets (OWA) | **0.841** | — | 0.972 | *0.840* | *0.821* |

| Dataset | TimeMixer | TimeMixer++ | SparseTSF | TimeMixer (paper) | TimeMixer++ (paper) |
|---|---|---|---|---|---|
| **PEMS (MAE ↓)** | | | | | |
| PEMS03 | **14.81** | — | 28.28 | — | — |
| PEMS04 | **19.22** | — | 36.12 | — | — |
| PEMS07 | **20.69** | — | 42.89 | — | — |
| PEMS08 | **15.33** | — | 29.25 | — | — |
| Average | **17.51** | — | 34.13 | *17.41* | *15.91* |
| **Imputation (MSE ↓)** | | | | | |
| ETTh1 | **0.1201** | — | 0.1633 | — | — |
| ETTh2 | **0.0664** | — | 0.0703 | — | — |
| ETTm1 | **0.0470** | — | 0.0715 | — | — |
| ETTm2 | **0.0278** | — | 0.0327 | — | — |
| Weather | **0.0315** | — | 0.0344 | — | — |
| Electricity | **0.0803** | — | 0.1304 | — | — |
| **Few-shot (MSE ↓)** | | | | | |
| ETTh1 | 0.6865 | — | **0.5517** | — | — |
| ETTh2 | 0.4550 | — | **0.4297** | — | — |
| ETTm1 | 0.6098 | — | **0.4935** | — | — |
| ETTm2 | 0.3018 | — | **0.2885** | — | — |
| ETT average | 0.5133 | — | **0.4408** | *0.4530* | *0.3960* |
| Weather | **0.2536** | — | 0.2708 | *0.2420* | *0.2410* |
| Electricity | **0.2093** | — | 0.2238 | *0.1870* | *0.1680* |
| **Zero-shot (MSE ↓)** | | | | | |
| ETTh1 → ETTh2 | 0.3669 | — | **0.3596** | *0.4270* | *0.3670* |
| ETTh1 → ETTm2 | 0.3268 | — | **0.3207** | *0.3610* | *0.3010* |
| ETTh2 → ETTh1 | 0.6387 | — | **0.4527** | *0.6790* | *0.5110* |
| ETTm1 → ETTh2 | 0.4640 | — | **0.3958** | *0.4520* | *0.4170* |
| ETTm1 → ETTm2 | 0.2998 | — | **0.2955** | *0.3290* | *0.2910* |
| ETTm2 → ETTm1 | 0.5917 | — | **0.5329** | *0.5540* | *0.4270* |
| **Anomaly (F1 % ↑)** | | | | | |
| PSM | **96.72** | — | 95.26 | — | *97.60* |
| SMD | **84.18** | — | 80.02 | — | *86.50* |
| MSL | 48.02 | — | **77.80** | — | *85.82* |
| SMAP | **73.05** | — | 66.76 | — | *73.10* |
| SWaT (not obtainable) | — | — | — | — | *94.64* |
| **Classification (accuracy % ↑)** | | | | | |
| EthanolConcentration | 27.00 | — | **29.28** | — | — |
| FaceDetection | **70.03** | — | 68.59 | — | — |
| Handwriting | 6.35 | — | **8.59** | — | — |
| Heartbeat | **75.61** | — | 54.63 | — | — |
| JapaneseVowels | 46.49 | — | **61.62** | — | — |
| PEMS-SF | 78.61 | — | **84.39** | — | — |
| SelfRegulationSCP1 | **90.78** | — | 82.25 | — | — |
| SelfRegulationSCP2 | **56.11** | — | 52.78 | — | — |

| Dataset | TimeMixer | TimeMixer++ | SparseTSF | TimeMixer (paper) | TimeMixer++ (paper) |
|---|---|---|---|---|---|
| SpokenArabicDigits | **95.54** | — | 94.04 | — | — |
| UWaveGestureLibrary | **77.19** | — | 72.81 | — | — |
| Average | **62.37** | — | 60.90 | — | *75.90* |

**Table 3.** Finished local runs per task, 11 October → 23 October 2026.

| Task | Protocol | TimeMixer: then → now | TimeMixer++: then → now | SparseTSF: then → now |
|---|---|---|---|---|
| Long-term forecasting | 8 datasets × 4 horizons = 32 | 30 → 30 | 21 → 21 | 20 → 32 |
| M4 | 6 subsets | 0 → 6 | published only | 6 → 6 |
| PEMS | 4 datasets | 0 → 4 | published only | 4 → 4 |
| Imputation | 6 datasets × 4 mask rates = 24 | 20 → 24 | published only | 24 → 24 |
| Few-shot | 6 datasets × 4 horizons = 24 | 0 → 24 | published only | 23 → 24 |
| Zero-shot | 6 pairs × 4 horizons = 24 | 0 → 24 | published only | 24 → 24 |
| Anomaly detection | 5 datasets | 0 → 4 | published only | 2 → 4 |
| Classification | 10 datasets | 0 → 10 | published only | 10 → 10 |

**Reading Table 2.**

- **SparseTSF matches or beats TimeMixer where its inductive bias fits.** It wins long-term forecasting on both hourly ETT datasets, few-shot on all four ETT datasets, and zero-shot on all six transfer pairs, with $10^1$–$10^3$ parameters against TimeMixer's $10^4$–$10^6$. With so few parameters there is little to overfit when data are scarce or the target distribution shifts.
- **SparseTSF loses most where signal crosses channels or is not periodic in the input window.** On PEMS (traffic sensor networks) TimeMixer's MAE is half of SparseTSF's (17.51 against 34.13), and on imputation TimeMixer wins all six datasets.
- **The local TimeMixer++ is undertrained** (Section 2.3): on ETTh1 its local MSE is 0.747 against 0.419 published. Its local column measures the reimplementation at a stable learning rate, not the architecture.
- **TimeMixer's anomaly F1 on MSL (48.0) is a failed run.** On the two repeated seeds it scores 80.2 and 79.4 (Section 4.2). Single-seed anomaly comparisons should not be read as rankings.

## 2.3 Why the comparison cannot be completed on this setup

Table 2 still has gaps and protocol deviations. Each has a stated cause:

1. **TimeMixer++ is not run per task.** No official code exists. The reimplementation diverges (NaN losses) at the paper's learning rates, and the coordinated stability fixes (skipping non-finite steps, clamped attention and FFT amplitudes) only hold at `lr = 5e-5` with step decay. At that rate it is undertrained, so per-task runs would mainly measure the instability. Its per-task columns are therefore published values only.
2. **Memory and time on a 4 GB GPU.**
    - TimeMixer on Traffic at horizons 336 and 720 does not fit, which leaves Traffic partial (2/4).
    - TimeMixer++ was not run on Electricity and Traffic, and finished only horizon 96 on Solar-Energy.
    - Test-time metrics for Electricity at horizon 720 and Traffic at 336/720 had to be accumulated batch by batch (an opt-in mode verified against the default to 6–7 significant digits).
3. **SWaT is not obtainable.** It is gated behind the iTrust request form. A third-party copy exists but carries no redistribution right and lacks the standard attack-period test split, so anomaly detection covers 4 of 5 datasets.
4. **Imputation input length.** The TimeMixer++ paper uses an input of 1,024 steps. SparseTSF needs its period to divide the input length, and 1,024 admits only powers of two, which cannot align with the 24-

hour cycle of hourly data. All local imputation runs therefore use 96 steps, and the published TimeMixer++ imputation numbers are not comparable (no published column).

5. **Configurations that do not exist.** TimeMixer publishes no anomaly-detection or classification settings; it runs there with its own long-term architecture, untuned. SparseTSF publishes no heads for imputation, anomaly detection or classification; the heads used here were designed in this study (Section 3.1).
6. **Early stopping on test** in three upstream training loops (imputation, anomaly, classification). This is kept, because changing it would invalidate every existing number, and those results are labelled a sensitivity analysis.
7. **One seed for most cells.** Only the headline comparisons were repeated (Section 4.2); TimeMixer was repeated on PEMS and anomaly detection only, because its imputation alone takes about ten hours per seed on this machine.
8. **Energy** is measured with CodeCarbon. GPU power is measured (NVML); CPU and RAM power are estimates from rated power, since Windows exposes no RAPL counters. Ratios between models measured in the same session are reliable; absolute values and cross-session comparisons are not (Section 4.4).

## 3. Ideas tested to improve the backbone

SparseTSFPlus is developed under fixed rules, so that each result can be attributed to one idea:

- **Off by default.** Every unit is off by default; with all units off the model is bitwise SparseTSF.
- **No dependence on the channel count.** No unit adds a parameter whose size depends on the number of channels, so a model trained on one dataset can be applied to another (the zero-shot task relies on this).
- **Validation decides.** Units are chosen on validation loss.
- **Prediction first.** Each idea's source, mechanism, prediction and refutation criterion are recorded in the progress log before its runs.

The ideas below are in the order they were tested. Table 4 summarises them.

**Table 4.** Ideas tested, with their outcome.

| # | Idea | Source | Extra parameters | Outcome |
|---|---|---|---|---|
| 3.1 | Heads for all task families | SparseTSF's own primitives | head only | Enabling step; SparseTSF on all 8 tasks |
| 3.2 | Phase mixing | TimeMixer++ dual-axis decomposition | $p^2$ | **Adopted**: PEMS −25%; overfits small sets |
| 3.3 | Iterative refinement | Imputation loss definition; iterative imputation | 0 | **Adopted**: largest single gain |
| 3.4 | Mask-aware aggregation | Partial convolutions | 0 | **Adopted** with refinement |
| 3.5 | Period selection; phase-aligned folding | SparseTSF's period; TimeMixer++ padding | 0 | Large lever (ETTh1 −3.2%, PEMS08 −24%) |
| 3.6 | Multi-period ensemble | TimeMixer multi-scale ensemble | one map per period | Rejected: inconsistent |
| 3.7 | Trend–season decomposition | Autoformer / TimeMixer | one map | **Adopted** (learned trend); flat trend rejected |
| 3.8 | Leave-one-out reconstruction | Noise2Self / Noise2Void | 0 | Rejected |
| 3.9 | Pooled classification readouts | x-vectors; piecewise aggregate approximation | smaller | Rejected (one dataset gains) |
| 3.10 | Instance normalisation (RevIN) | Kim et al. 2022 | 0 | Rejected: provably no effect |
| 3.11 | One fixed configuration | Thesis goal; TimeMixer++ design | — | **Headline model** |
| 3.12 | Learned gates | ReZero | 1 per unit | Refinement gate **adopted**; phase gate rejected |
| 3.13 | Phase mixing on missing values | Partial convolutions | 0 | Mask-aware: candidate, unconfirmed; complete-only rejected |

| # | Idea | Source | Extra parameters | Outcome |
|---|---|---|---|---|
| 3.14 | Channel context | SOFTS; iTransformer | 2–9 | Correlation: confirmed on PEMS (−5.3%), not yet in the full suite |
| 3.15 | Period by FFT | TimesNet; TimeMixer++ | 0 | Rejected without training |
| 3.16 | Seed repeats | Standard practice | — | Confirmed the headline; corrected three claims |

### 3.1 Extending SparseTSF to every task family

- **Why.** SparseTSF was published for long-term forecasting only. A general-purpose comparison needs it on all eight tasks before anything can be improved.
- **Where from.** SparseTSF's own components: the aggregating convolution and the cross-period linear map.
- **What.**
  - **Reconstruction** (imputation and anomaly detection) folds the input and maps periods back to periods (a square linear map), instead of input periods to future periods.
  - **Classification** aggregates with the convolution, flattens, and applies one linear layer.
  - Forecasting delegates to the untouched original code, so published SparseTSF numbers are reproduced exactly.
- **Result.** SparseTSF runs on all eight tasks (Tables 1 and 2). The classification head is the only place it is not small, because its size scales with channels × input length × classes (up to 970,712 parameters).
- **Lesson.** The per-task gaps in Table 1 had distinct causes, which set the agenda for the units below:
  - PEMS: a channel-independent model with an eight-weight head;
  - imputation: a reconstruction that cannot tell a missing value from an observed zero;
  - classification: a flatten readout.

### 3.2 Phase mixing

- **Why.** SparseTSF's forecast map, written per channel, is a Kronecker product `I_p` ⊗ `W_n`: one small map across periods, applied independently at each of the `p` phases. The shape of the signal *within* a period is never modelled.
- **Where from.** TimeMixer++'s dual-axis "time image" decomposition, which attends along both the period axis and the phase axis. It is reduced here to one shared linear map along the phase axis.
- **What.** A second linear map across phases turns the forecast into `W_p` ⊗ `W_n`, a factorised full map. It costs `p²` parameters (16 at p = 4, 576 at p = 24) independent of input length, against `seq_len × pred_len` for a dense map. It is zero-initialised, so training starts from SparseTSF.
- **Result.**
  - **PEMS:** at period 4 it cut the MAE on all four datasets from 34.13 to 25.50 (−25%) with 93 parameters, within 0.6% of a dense map with 12× the parameters.
  - **ETTh1:** it overfits (+2.7%, worse on validation as well). A rank-4 factorisation removes most of the overfit (0.394 → 0.387) but never beats the plain model.
  - **Imputation:** over three seeds its effect is within noise (Section 3.16).
- **Lesson.** Within-period shape is the main thing SparseTSF misses on traffic data, and a factorised map recovers nearly all of a dense model's accuracy at a small fraction of its size. Its weakness is small datasets.

### 3.3 Iterative refinement

- **Why.** Imputation is scored only on missing positions. At a mask rate of 0.5, half of the convolution's window is literal zeros that the model cannot distinguish from observations, and SparseTSF's error grew monotonically with the mask rate (ETTm1 +52% behind TimeMixer on average).

- **Where from.** The loss definition itself, and the long tradition of iterative imputation (for example chained equations, `mice`). The refinement step is the cheapest form of it.
- **What.** Run the model once, fill the holes with its estimate, recompute the normalisation on the now complete series, and run the *same* weights again. No new parameters; one extra forward pass.
- **Result.** It was the largest single gain in the study. On ETTh1 imputation it cut the error by 26% (0.1633 → 0.1207) with no added parameter; the six-dataset mean fell from 0.0838 to 0.0733. On ETTh2, without the gate of Section 3.12, it made results worse (0.0703 → 0.0955).
- **Lesson.** The most effective unit costs nothing; it changes how the existing model is used. It needed a safe default before it could be on everywhere.

### 3.4 Mask-aware aggregation

- **Why.** Same diagnosis as Section 3.3: missing values enter the aggregating convolution as zeros.
- **Where from.** Partial convolutions for image inpainting (Liu et al., ECCV 2018).
- **What.** Compute the convolution from observed values only, and rescale by the share of the window that was observed.
- **Result.**
  - **Alone:** no effect (ETTh1, mask 0.5: 0.2141 → 0.2136).
  - **With refinement:** it helps on large datasets (ETTm1: 0.0497 → 0.0426) and hurts slightly on small ones (ETTh1: 0.1207 → 0.1263).
- **Lesson.** The two missing-value units are complementary, and how much capacity pays off depends on the training-set size (about 34,000 windows for ETTm1 against 8,500 for ETTh1).

### 3.5 Period selection and phase-aligned folding

- **Why.** SparseTSF's period is a fixed hyperparameter from its paper (24 for hourly data, 4 for 15- and 10-minute data, 12 for PEMS). It decides how many segments the model sees, which is all of its capacity.
- **Where from.**
  - SparseTSF's own period hyperparameter, here chosen on validation instead of taken from the paper.
  - TimeMixer++'s pad-and-fold, corrected so that every forecast step keeps its phase.
- **What.**
  - A sweep of periods, chosen on validation.
  - Padding at the front of the window, so periods that do not divide the input length become usable without shifting the phase of any forecast step.
- **Result.**
  - **ETTh1:** prefers period 12 to the published 24 (−3.2% MSE at horizon 96), which turns a 2.8% loss against TimeMixer into a 0.5% win.
  - **PEMS08:** prefers period 1, no folding at all (−24% MAE).
  - **Weather and ETTm1:** insensitive to the period.
- **Lesson.** The period is a large lever, and shorter periods (more segments per window) tend to help. The fixed configuration still uses the published periods, because choosing per dataset would be tuning; Section 3.15 tests an automatic rule.

### 3.6 Multi-period ensemble

- **Why.** One hand-chosen period is a strong assumption about which cycle dominates.
- **Where from.** TimeMixer's ensemble over temporal scales, applied to periods instead of resolutions.
- **What.** Run the fold at several periods and add the branches behind zero-initialised gates.
- **Result.** Small and inconsistent: −1.8% on ETTm1, +0.7% on ETTh1, +0.5% on Weather. It never had the lowest validation loss.

- **Lesson.** One well-chosen period already captures what extra periods add here. Rejected.

### 3.7 Trend–season decomposition

- **Why.** Drift over the input window is something a period-wise map extrapolates poorly.
- **Where from.** The moving-average decomposition of Autoformer (Wu et al., NeurIPS 2021), also used by TimeMixer.
- **What.** Split the input into a moving-average trend and a seasonal remainder; the seasonal part goes through the period map. Two trend treatments were tested:
  - holding the trend at its last value;
  - giving the trend its own period map.
- **Result.**
  - **Flat trend:** harmful on every dataset (+13% to +27% MSE). It removes drift that SparseTSF could already model.
  - **Learned trend:** the best forecasting unit on validation on all three datasets screened. It improves ETTm1 at all horizons (−0.8% to −2.7%) and Weather (−0.2% to −1.0%), is mixed on ETTh1, and doubles the period map (581 → 1,157 parameters on ETTm1).
- **Lesson.** A decomposition helps only if the trend gets its own predictor. The gain is small, but over three seeds the configuration that contains it beats SparseTSF on long-term forecasting (Section 4.2).

### 3.8 Leave-one-out reconstruction

- **Why.** On SMD, SparseTSF's anomaly detector was limited by recall (73.9) rather than precision (87.3). A reconstruction model can represent the identity map exactly, and copying the input reconstructs anomalies faithfully too.
- **Where from.** Blind-spot denoising (Noise2Void, Krull et al., CVPR 2019; Noise2Self, Batson and Royer, ICML 2019): predict each point without seeing it.
- **What.** Remove the self-term of the reconstruction map.
- **Result.** SMD recall improved slightly (73.90 → 75.39; F1 80.02 → 80.55), but PSM got worse (F1 95.26 → 93.23).
- **Lesson.** Copying the input explains little of SMD's gap. Rejected.

### 3.9 Pooled classification readouts

- **Why.** The flatten readout scales with input length (up to 970,712 parameters) and has no invariance to time shifts.
- **Where from.**
  - Statistics pooling in speaker recognition (x-vectors, Snyder et al., ICASSP 2018).
  - Segment means as in piecewise aggregate approximation (Keogh et al., 2001).
- **What.** Four variants:
  - per-channel mean, standard deviation and maximum over time;
  - segment means, which keep coarse time order;
  - a wider aggregator;
  - a fix for a padding bias in the centring step.
- **Result.** The ten-dataset average fell from 60.90 to 36.69 with global statistics pooling. Shape- driven datasets collapsed (UWaveGestureLibrary 72.8 → 9.4), and segment pooling did not recover them. One dataset gained strongly: Heartbeat 54.6 → 71.7 with a readout 134× smaller, where the flatten readout had a 242:1 parameter-to-sample ratio.
- **Lesson.** Pooling helps where classes differ in amplitude statistics and fails where they differ in shape. The classification gap to TimeMixer++ (15 points, published) lies in what the backbone extracts, not in the

readout. Rejected as a default.

### 3.10 Instance normalisation (RevIN)

- **Why.** RevIN (Kim et al., ICLR 2022) is a standard remedy for distribution shift in forecasting.
- **What.** Divide each window by its standard deviation and multiply the forecast back.
- **Result.** It provably has no effect on SparseTSF. Its forecast map is linear and bias-free, hence `f(x/σ)·σ = f(x)`; measured, the outputs differ by $1.5 \times 10^{-5}$ (floating-point rounding).
- **Lesson.** Any per-window rescaling is inert on SparseTSF. A whole family of fixes can be ruled out without training. Rejected.

### 3.11 One fixed configuration across all tasks

- **Why.** Choosing units per dataset is an ablation, not a model. TimeMixer++ is one architecture with thin task heads, and the thesis target is the same for SparseTSFPlus.
- **Where from.** The thesis goal, and how TimeMixer++ itself is built.
- **What.** One configuration for every task and dataset:
    - learned-trend decomposition and phase mixing in the backbone;
    - gated refinement and mask-aware aggregation whenever the input has missing values;
    - the standard heads, with SparseTSF's published period everywhere.

  The **per-task** version, which uses each task's best unit set, is reported alongside it to measure the cost of generality.
- **Result.** Over three seeds (Section 4.2):
    - **Against SparseTSF,** the fixed configuration is better beyond noise on PEMS, imputation, M4, few-shot and long-term forecasting.
    - **Against the per-task version,** it is within 0.7% on every forecasting task; it is the better of the two on long-term forecasting.
    - **Energy** is 1.35× SparseTSF's.
- **Lesson.** One model costs almost nothing against per-task tuning. The remaining gaps to TimeMixer are structural: cross-channel structure (PEMS), an input window shorter than some daily cycles (Weather, Solar), and classification.

### 3.12 Learned gates (ReZero)

- **Why.** In the first one-model run, two units were not safe by default: refinement on ETTh2 and phase mixing on Electricity imputation. Both raised the *training* loss as well, so the harm was visible to training rather than being overfitting.
- **Where from.** ReZero (Bachlechner et al., UAI 2021): a residual branch multiplied by one scalar initialised at zero. The multi-period ensemble already used such gates.
- **What.** Each gate is one learned number per unit, starting at zero, so the model starts exactly as if the unit were off (verified bitwise).
    - **Refinement:** the output becomes pass 1 + g·(pass 2 − pass 1).
    - **Phase mixing:** the phase image becomes x + g·W_p·x.
- **Result.**

    - **Refinement gate:** the six-dataset imputation mean fell from 0.0716 to 0.0651, better on all six datasets. On ETTh2 the gate settled at about 0.5 (averaging the two passes) and fixed the failure (0.0955 → 0.0641). It settled near 1 on the larger datasets.
    - **Phase gate:** no effect (0.0688 against 0.0689).
- **Lesson.** Letting a unit be partly on can repair it, and the learned values are interpretable. A gate cannot block harm that training actively moves into. The ungated comparison was not repeated over seeds.

### 3.13 Phase mixing on inputs with missing values

- **Why.** After the refinement gate, phase mixing looked like the only source of the imputation gap between the fixed and per-task versions, with harm only at high mask rates (0.375 and 0.5). That pattern suggested phase mixing spreads the holes along the phase axis.
- **Where from.** Partial convolutions again (Section 3.4).
- **What.** Two forms:
    - **Complete inputs only:** skip phase mixing in the first pass, which still has the holes.
    - **Mask-aware:** mix only observed positions and rescale.
- **Result.**
    - **Complete inputs only:** rejected (+23–42% on Electricity, +29–39% on ETTh1, even at low mask rates).
    - **Mask-aware:** on one seed it brought the fixed imputation form within 1.0% of the form without phase mixing. Over three seeds the difference is within noise (−2.3% ± 2.1% on ETTh1/ETTh2, −6.0% ± 7.4% on Electricity). Its one consistent effect: on Electricity at mask 0.5 it is the best variant on all three seeds.
- **Lesson.** The first pass needs phase mixing. More importantly, the problem this idea targeted turned out to be largely run-to-run noise (Section 3.16). Kept as an unconfirmed candidate.

### 3.14 Channel context for PEMS

- **Why.** PEMS is the largest remaining gap to TimeMixer. Neighbouring traffic sensors move together, SparseTSF treats every channel independently, and TimeMixer runs PEMS with its channels embedded jointly.
- **Where from.**
    - The aggregate-and-redistribute core of SOFTS (Han et al., NeurIPS 2024).
    - Attention across variates in iTransformer (Liu et al., ICLR 2024), with its learned projections removed.
- **What.** Both forms add a gated cross-channel term to each forecast, without any parameter that depends on the channel count:
    - **Channel mean:** the mean of all channels gets its own small map, added to each channel in proportion to that channel's loading on the mean.
    - **Channel correlation:** each channel adds a softmax-weighted mix of the other channels' forecasts, weighted by input correlation, with a learned temperature (2 parameters).
- **Result.**
    - **One seed:** PEMS MAE −3.9% (mean) and −4.9% (correlation), with no harm on the long-term guard datasets. Neither reached the pre-registered 5% validation threshold, so neither was run on the full suite.
    - **Three seeds:** both gains are findings (mean −3.1% ± 0.7%; correlation −5.25% ± 0.29%).
    - PEMS remains about 48% behind TimeMixer.
- **Lesson.** Cross-channel information helps a little at almost no cost, but TimeMixer's PEMS lead needs deeper channel interaction. A screening threshold applied to one seed rejected a small effect that proved real. Channel correlation is a candidate awaiting a full-suite run.

### 3.15 Choosing the period by FFT

- **Why.** Section 3.5 showed that the period matters, but choosing it per dataset is tuning. A fixed rule computed from the data would keep the model general.
- **Where from.** TimesNet (Wu et al., ICLR 2023) and TimeMixer++ take periods from the strongest FFT frequencies.
- **What.** On each training split, take the strongest period between 2 and 48 steps (at least two periods in the input), snapped to a divisor of 96.
- **Result.** The rule picks the published 24 on ETTh1 and ETTh2, and 48 on every other dataset. The 48 is the edge of the allowed band, because the spectra are dominated by slow variation. The period sweep had already measured 48 as worse (+6.9% on ETTm1, +7.7% on Weather), so no training was needed.
- **Lesson.** The strongest cycle in the data is the wrong target for SparseTSF, which benefits from short periods with more segments per window. Rejected.

### 3.16 Seed repeats

- **Why.** Most gains were a few percent or less, measured on one seed.
- **What.** Seeds 2022 and 2023 for every headline comparison, each run identical to the seed-2021 run except for the seed: about 600 runs.
- **Result.**
  - **Forecasting is very stable** (0.1–0.5% spread between seeds), so even gains of 1% are findings.
  - **Imputation varies by 4–7% between seeds,** more than the differences between SparseTSFPlus variants.
  - **TimeMixer's seed-2021 MSL run had failed.**
- **Lesson.** Three earlier claims were withdrawn or weakened:
  - the imputation cost of phase mixing;
  - the mask-aware fix;
  - SparseTSFPlus beating TimeMixer on anomaly detection.

  Future imputation comparisons need paired masks (every variant seeing the same holes) or more seeds.

## 4. Full comparison including SparseTSFPlus

### 4.1 Per dataset (seed 2021)

Table 5 extends Table 2 with the two SparseTSFPlus versions. The **fixed** column is the one-model configuration of Section 3.11. The **per task** column uses each task's best unit set, applied unchanged to all of that task's datasets. On anomaly detection and classification the per-task version has no units, so it equals SparseTSF.

**Table 5.** All four models (seed 2021).

| Dataset | TimeMixer | TimeMixer++ | SparseTSF | SparseTSFPlus fixed | SparseTSFPlus per task | TimeMixer (paper) | TimeMixer++ (paper) |
|---|---|---|---|---|---|---|---|
| **Long-term (MSE ↓)** | | | | | | | |
| ETTh1 | 0.4578 | 0.7467 | 0.4347 | **0.4291** | 0.4318 | *0.4465* | *0.4185* |
| ETTh2 | 0.3730 | 0.4507 | 0.3566 | **0.3509** | 0.3551 | *0.3648* | *0.3390* |

| Dataset | TimeMixer | TimeMixer++ | SparseTSF | SparseTSFPlus fixed | SparseTSFPlus per task | TimeMixer (paper) | TimeMixer++ (paper) |
|---|---|---|---|---|---|---|---|
| ETTm1 | **0.3841** | 0.4579 | 0.4172 | 0.4132 | 0.4109 | *0.3812* | *0.3685* |
| ETTm2 | **0.2799** | 0.3125 | 0.2884 | 0.2868 | 0.2866 | *0.2752* | *0.2687* |
| Weather | **0.2454** | 0.2756 | 0.2741 | 0.2727 | 0.2726 | *0.2403* | *0.2263* |
| Solar-Energy | **0.2319** | partial (1/4) | 0.3858 | 0.3649 | 0.3643 | *0.2162* | *0.2032* |
| Electricity | **0.1830** | — | 0.2218 | 0.2189 | 0.2198 | *0.1822* | *0.1645* |
| Traffic | partial (2/4) | — | 0.6346 | **0.6329** | 0.6349 | *0.4848* | *0.4158* |
| **M4 (OWA ↓)** | | | | | | | |
| M4, 6 subsets (OWA) | **0.841** | — | 0.972 | 0.914 | 0.918 | *0.840* | *0.821* |
| **PEMS (MAE ↓)** | | | | | | | |
| PEMS03 | **14.81** | — | 28.28 | 21.25 | 21.34 | — | — |
| PEMS04 | **19.22** | — | 36.12 | 27.37 | 27.43 | — | — |
| PEMS07 | **20.69** | — | 42.89 | 30.94 | 31.02 | — | — |
| PEMS08 | **15.33** | — | 29.25 | 22.24 | 22.22 | — | — |
| Average | **17.51** | — | 34.13 | 25.45 | 25.50 | *17.41* | *15.91* |
| **Imputation (MSE ↓)** | | | | | | | |
| ETTh1 | **0.1201** | — | 0.1633 | 0.1222 | 0.1230 | — | — |
| ETTh2 | 0.0664 | — | 0.0703 | 0.0770 | **0.0641** | — | — |
| ETTm1 | 0.0470 | — | 0.0715 | **0.0415** | 0.0420 | — | — |
| ETTm2 | 0.0278 | — | 0.0327 | **0.0268** | 0.0269 | — | — |
| Weather | 0.0315 | — | 0.0344 | 0.0300 | **0.0299** | — | — |
| Electricity | **0.0803** | — | 0.1304 | 0.1158 | 0.1048 | — | — |
| **Few-shot (MSE ↓)** | | | | | | | |
| ETTh1 | 0.6865 | — | 0.5517 | **0.5251** | 0.5337 | — | — |
| ETTh2 | 0.4550 | — | 0.4297 | 0.4173 | **0.4169** | — | — |
| ETTm1 | 0.6098 | — | 0.4935 | 0.4884 | **0.4850** | — | — |
| ETTm2 | 0.3018 | — | 0.2885 | 0.2867 | **0.2867** | — | — |
| ETT average | 0.5133 | — | 0.4408 | **0.4294** | 0.4306 | *0.4530* | *0.3960* |
| Weather | **0.2536** | — | 0.2708 | 0.2691 | 0.2686 | *0.2420* | *0.2410* |
| Electricity | **0.2093** | — | 0.2238 | 0.2259 | 0.2217 | *0.1870* | *0.1680* |
| **Zero-shot (MSE ↓)** | | | | | | | |
| ETTh1 → ETTh2 | 0.3669 | — | 0.3596 | **0.3554** | 0.3561 | *0.4270* | *0.3670* |
| ETTh1 → ETTm2 | 0.3268 | — | 0.3207 | **0.3184** | 0.3196 | *0.3610* | *0.3010* |
| ETTh2 → ETTh1 | 0.6387 | — | **0.4527** | 0.4629 | 0.4544 | *0.6790* | *0.5110* |
| ETTm1 → ETTh2 | 0.4640 | — | 0.3958 | 0.3944 | **0.3939** | *0.4520* | *0.4170* |
| ETTm1 → ETTm2 | 0.2998 | — | 0.2955 | **0.2938** | 0.2939 | *0.3290* | *0.2910* |
| ETTm2 → ETTm1 | 0.5917 | — | 0.5329 | **0.5223** | 0.5254 | *0.5540* | *0.4270* |
| **Anomaly (F1 % ↑)** | | | | | | | |
| PSM | **96.72** | — | 95.26 | 95.38 | 95.26 | — | *97.60* |
| SMD | **84.18** | — | 80.02 | 80.03 | 80.02 | — | *86.50* |
| MSL | 48.02 | — | **77.80** | 77.79 | 77.80 | — | *85.82* |
| SMAP | **73.05** | — | 66.76 | 66.74 | 66.76 | — | *73.10* |
| SWaT (not obtainable) | — | — | — | — | — | — | *94.64* |

| Dataset | TimeMixer | TimeMixer++ | SparseTSF | SparseTSFPlus fixed | SparseTSFPlus per task | TimeMixer (paper) | TimeMixer++ (paper) |
|---|---|---|---|---|---|---|---|
| **Classification (accuracy % ↑)** | | | | | | | |
| EthanolConcentration | 27.00 | — | **29.28** | 29.28 | 29.28 | — | — |
| FaceDetection | **70.03** | — | 68.59 | 68.59 | 68.59 | — | — |
| Handwriting | 6.35 | — | **8.59** | 8.59 | 8.59 | — | — |
| Heartbeat | **75.61** | — | 54.63 | 54.63 | 54.63 | — | — |
| JapaneseVowels | 46.49 | — | **61.62** | 61.62 | 61.62 | — | — |
| PEMS-SF | 78.61 | — | **84.39** | 84.39 | 84.39 | — | — |
| SelfRegulationSCP1 | **90.78** | — | 82.25 | 82.25 | 82.25 | — | — |
| SelfRegulationSCP2 | **56.11** | — | 52.78 | 52.78 | 52.78 | — | — |
| SpokenArabicDigits | **95.54** | — | 94.04 | 94.04 | 94.04 | — | — |
| UWaveGestureLibrary | **77.19** | — | 72.81 | 72.81 | 72.81 | — | — |
| Average | **62.37** | — | 60.90 | 60.90 | 60.90 | — | *75.90* |

## 4.2 Over three seeds

Table 6 gives the mean ± standard deviation over seeds 2021–2023 of each task's headline number, for every configuration that was repeated. The comparisons that are findings under the rule of Section 1.3:

- **Fixed vs SparseTSF:** PEMS −24.4% ± 0.9%, imputation (ETTh1/ETTh2) −16.2% ± 2.6%, M4 −6.1% ± 0.6%, few-shot −1.4% ± 0.5%, long-term −1.0% ± 0.2%.
- **Fixed vs per task:** long-term −0.4% ± 0.1% (fixed better); zero-shot +0.2% ± 0.1% (per task better).
- **TimeMixer vs fixed:** PEMS, by 47.7% ± 2.0%.

Not findings:

- **Fixed vs SparseTSF:** zero-shot (−0.42% ± 0.24%) and anomaly detection (+0.02 F1).
- **Fixed vs TimeMixer on anomaly detection:** +4.5, −2.5 and −2.9 F1 by seed.
- **Every imputation-variant difference** between SparseTSFPlus forms.

**Table 6.** Mean ± standard deviation over seeds 2021, 2022 and 2023. The per-task imputation form is gated refinement; "—" means not repeated.

| Task | SparseTSF | SparseTSFPlus fixed | SparseTSFPlus per task | TimeMixer |
|---|---|---|---|---|
| Long-term (ETT×4, Weather; 4 horizons) (MSE ↓) | 0.3538 ± 0.0007 | 0.3503 ± 0.0003 | 0.3516 ± 0.0002 | — |
| M4 (6 subsets) (OWA ↓) | 0.973 ± 0.004 | 0.914 ± 0.002 | 0.921 ± 0.005 | — |
| PEMS (4 datasets) (MAE ↓) | 34.12 ± 0.01 | 25.79 ± 0.30 | — | 17.46 ± 0.06 |
| Few-shot (6 datasets) (MSE ↓) | 0.3747 ± 0.0014 | 0.3694 ± 0.0007 | 0.3707 ± 0.0020 | — |
| Zero-shot (6 pairs) (MSE ↓) | 0.3931 ± 0.0011 | 0.3914 ± 0.0003 | 0.3905 ± 0.0003 | — |
| Anomaly (4 datasets) (F1 ↑) | 80.47 ± 0.85 | 80.49 ± 0.53 | — | 80.81 ± 4.61 |
| Imputation (ETTh1, ETTh2; 4 mask rates) (MSE ↓) | 0.1151 ± 0.0015 | 0.0965 ± 0.0035 | 0.0947 ± 0.0012 | — |
| Imputation (Electricity; 4 mask rates) (MSE ↓) | — | 0.1067 ± 0.0079 | 0.1007 ± 0.0044 | — |

## 4.3 Model size

**Table 7.** Trainable parameters (range over datasets and horizons).

| Task | SparseTSF | SparseTSFPlus (fixed) | TimeMixer | TimeMixer++ |
|---|---|---|---|---|
| Long-term forecasting (seq_len 96) | 41–4,325 | 633–8,661 | 75,497–2,511,077 | 112,951–233,885 |
| Short-term forecasting, M4 | 13–33 | 37–617 | 10,525–100,069 | — |
| Short-term forecasting, PEMS | 21 | 173 | 869,170–1,237,791 | — |
| Imputation | 41–581 | 598–618 | 57,833–60,345 | — |
| Anomaly detection | 41 | 666 | 62,573–62,813 | — |
| Classification | 3,142–970,712 | 3,142–970,712 | 14,349–18,227,514 | — |

### 4.4 Energy

The matched energy protocol trains every model with the same settings: 5 epochs, no early stopping, batch 64. It runs long-term forecasting on five datasets, reporting the total over four horizons in kWh. Its accuracy is not comparable with the tables above. Short runs vary between sessions (the SparseTSF total measured in July was 42% higher than in October), so SparseTSF, SparseTSFPlus and TimeMixer were measured in one session. TimeMixer++ comes from an earlier session and is indicative only.

**Table 8.** Energy per dataset (kWh, total over four horizons).

| Dataset | SparseTSF | SparseTSFPlus (fixed) | SparseTSFPlus (per task) | TimeMixer | TimeMixer++ |
|---|---|---|---|---|---|
| ETTh1 | 0.00037 | 0.00040 | 0.00041 | 0.00425 | 0.01176 |
| ETTh2 | 0.00038 | 0.00041 | 0.00041 | 0.00440 | 0.01024 |
| ETTm1 | 0.00104 | 0.00138 | 0.00136 | 0.01865 | 0.04026 |
| ETTm2 | 0.00109 | 0.00139 | 0.00138 | 0.01879 | 0.03825 |
| Weather | 0.00162 | 0.00248 | 0.00242 | 0.03903 | 0.04891 |
| **Total** | **0.00450** | **0.00606** | **0.00598** | **0.08511** | **0.14941** |
| *× SparseTSF* | *1.00* | *1.35* | *1.33* | *18.91* | *33.19* |
| *Measured* | *2026-10-21* | *2026-10-21* | *2026-10-21* | *2026-10-21* | *2026-07-23* |

### 4.5 Discussion

The one-model SparseTSFPlus answers half of the thesis question. It is efficient: $10^1$–$10^3$ parameters outside classification, and 1.35× SparseTSF's energy, about 7% of TimeMixer's. With one configuration it improves on SparseTSF beyond run-to-run noise on five of the eight tasks, and choosing units per task adds nothing measurable on forecasting. It leads TimeMixer on few-shot and zero-shot forecasting, where small models generalise better. It is not yet as capable as TimeMixer on three fronts, each with a known cause:

- **PEMS and other cross-channel data:** the model is channel-independent by design. Channel context recovers about 5%, against a gap of about 48%.
- **Long input dependence:** Weather and Solar have daily cycles longer than the 96-step input, which no period-based mechanism can recover.
- **Classification:** neither SparseTSF's backbone nor any readout tested reaches TimeMixer++'s published accuracy.

## 5. Conclusions and next steps

1. **One efficient model is feasible.** A fixed SparseTSFPlus configuration of at most 8,661 parameters outside classification improves on SparseTSF on five tasks beyond seed noise, at no measurable cost against per-task tuning.

2. **The cheapest units did the most:** iterative refinement, its one-parameter gate, and phase mixing. Several ideas from larger models did not transfer: RevIN, FFT period choice, the multi-period ensemble, leave-one-out reconstruction, and pooled readouts.
3. **Seed repeats are necessary on imputation and anomaly detection.** Three single-seed claims did not survive them.

**Decisions for discussion.**

- Run channel correlation on the full suite. It is confirmed on PEMS over seeds, but was held back by the one-seed screen.
- Make imputation comparisons paired (the same masks for every variant).
- Decide how to address cross-channel structure inside the backbone without breaking transfer between datasets.

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## Appendix: reproducing this report

All runners are idempotent: a configuration whose log already holds its result is skipped. The tables are regenerated, and the PDF rebuilt (pandoc and Chrome or Edge), by:

```
cd TimeMixer
venv/Scripts/python.exe benchmarks/tools/build_progress_document.py --pdf
```

Detailed per-task reports are in `TimeMixer/reports/`:

- `results_comparison_<task>.md` for each task's comparison;
- `results_sparsetsf_plus_report.md` for the units;
- `results_sparsetsf_plus_suite.md` for the one-model runs;
- `results_seed_repeats.md` for the seed repeats.

The dated record of every step, with its plan and prediction written before its runs, is `reports/progress_log.md`.